



SQL ENGINEERING HANDBOOK

Official Documentation

Authoritative references for engines and tools

Documentation

Part of the [Resources Library](#) — [SQL Engineering Handbook](#)

This is the one file in the library where "why is this here" barely needs an answer — it's the official word, straight from whoever builds the thing. When a module in this repo and a database's own documentation disagree, trust the documentation. Open an issue on the module.

[!NOTE] Root documentation URLs are about as stable as links get — vendors rarely move their main docs domain. If one of these has moved by the time you're reading this, it's a fast PR: see [Contribution Guidelines](#).

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Relational Databases

Recommended reading order: pick the dialect you're actually using and read that one's tutorial/getting-started section end to end before treating any of these as a reference. Don't read all six front to back — that's what books .md → *SQL Pocket Guide* is for.

PostgreSQL Official Documentation

Field	Details
Author / Organization	The PostgreSQL Global Development Group
Category	Relational Databases
Difficulty	Beginner (Tutorial) → Advanced (full manual)
Best For	The most complete, precisely written open-source database documentation available — the standard other docs get measured against

Field	Details
Why It Is Recommended	Written by the people who implement the features, updated on every release, and unusually exact about edge-case behavior (NULL handling, MVCC visibility rules) that third-party tutorials tend to gloss over
Key Topics Covered	SQL syntax and functions, indexing (B-tree, GIN, GiST, BRIN), MVCC and transaction isolation, extensions, EXPLAIN and query planning
Estimated Time	2–3 hours for the Tutorial section; ongoing as reference after that
Official Website	postgresql.org/docs/
Free or Paid	Free
Prerequisites	None for the Tutorial; 03_JOINS for the rest
Who Should Read It	Anyone using PostgreSQL, from first query to EXPLAIN ANALYZE
Related SQL Handbook Modules	Applies across all modules if PostgreSQL is your dialect

MySQL Official Documentation

Field	Details
Author / Organization	Oracle Corporation
Category	Relational Databases
Difficulty	Beginner (Tutorial) → Advanced (Reference Manual)
Best For	Authoritative InnoDB and replication behavior, and MySQL-specific syntax variations
Why It Is Recommended	The only fully authoritative source for InnoDB locking behavior and replication semantics, which differ from PostgreSQL in ways that matter for correctness, not just syntax
Key Topics Covered	SQL syntax, InnoDB storage engine, replication, indexing, EXPLAIN
Estimated Time	2 hours for Tutorial; ongoing as reference
Official Website	dev.mysql.com/doc/
Free or Paid	Free
Prerequisites	None for the Tutorial
Who Should Read It	Anyone using MySQL or MariaDB in production

Field	Details
Related SQL Handbook Modules	Applies across all modules if MySQL is your dialect

Microsoft Learn — SQL Server & Azure SQL

Field	Details
Author / Organization	Microsoft
Category	Relational Databases
Difficulty	Beginner → Advanced
Best For	T-SQL specifics and the Azure SQL / Fabric platform ecosystem around SQL Server
Why It Is Recommended	Microsoft Learn consolidates docs, structured learning paths, and certification prep in one place — unusually good for going from zero to a specific credential
Key Topics Covered	T-SQL syntax, Intelligent Query Processing, indexing, Azure SQL Database, SQL Server on Linux
Estimated Time	2–3 hours for the "Get started" learning path
Official Website	learn.microsoft.com/en-us/sql/
Free or Paid	Free
Prerequisites	None
Who Should Read It	Anyone on the Microsoft data stack, or preparing for a Microsoft data certification
Related SQL Handbook Modules	Applies across all modules if T-SQL is your dialect

Oracle Database SQL Documentation

Field	Details
Author / Organization	Oracle Corporation
Category	Relational Databases
Difficulty	Intermediate → Advanced
Best For	Enterprise Oracle Database features — PL/SQL, partitioning, RAC — used in large legacy enterprise systems

Field	Details
Why It Is Recommended	The only authoritative source for Oracle-specific SQL extensions (hierarchical queries with <code>CONNECT BY</code> , analytic function variations) that diverge meaningfully from standard SQL
Key Topics Covered	SQL and PL/SQL reference, partitioning, Real Application Clusters (RAC), Autonomous Database
Estimated Time	Reference-style; budget 3–4 hours for the SQL Language Reference basics
Official Website	docs.oracle.com/en/database/oracle/oracle-database/
Free or Paid	Free
Prerequisites	03_JOINS, 05_SUBQUERIES
Who Should Read It	Engineers in Oracle-standardized enterprise environments
Related SQL Handbook Modules	Applies across all modules if Oracle is your dialect

SQLite Documentation

Field	Details
Author / Organization	The SQLite Consortium
Category	Relational Databases
Difficulty	Beginner → Intermediate
Best For	Understanding SQLite's intentionally different type system and its query planner, for embedded/local use cases
Why It Is Recommended	Unusually short and precisely written for a database manual — SQLite's own documentation is a good model of technical writing regardless of whether you use the database daily
Key Topics Covered	Dynamic typing rules, <code>EXPLAIN QUERY PLAN</code> , file format, <code>WITHOUT ROWID</code> tables
Estimated Time	1–2 hours
Official Website	sqlite.org/docs.html
Free or Paid	Free
Prerequisites	01_FUNDAMENTALS
Who Should Read It	Mobile/embedded developers, and anyone prototyping locally before deploying to a client-server database

Field	Details
Related SQL Handbook Modules	01_FUNDAMENTALS, 15_INDEXES

MariaDB Documentation (Knowledge Base)

Field	Details
Author / Organization	MariaDB Foundation / MariaDB plc
Category	Relational Databases
Difficulty	Beginner → Advanced
Best For	Where MariaDB has diverged from MySQL since their fork — storage engines, JSON handling, and features MySQL doesn't have
Why It Is Recommended	Written specifically to flag divergence from MySQL, which is exactly the information a MySQL-background reader needs and won't get from MySQL's own docs
Key Topics Covered	Storage engines (Aria, ColumnStore), JSON functions, MariaDB-specific replication (Galera Cluster)
Estimated Time	1–2 hours if you already know MySQL; longer otherwise
Official Website	mariadb.com/kb/en/
Free or Paid	Free
Prerequisites	MySQL Official Documentation, for comparison
Who Should Read It	Anyone migrating from or to MariaDB from MySQL
Related SQL Handbook Modules	Applies across all modules if MariaDB is your dialect

Cloud Data Warehouses

Recommended reading order: whichever platform your employer or project already uses. If choosing freely, Snowflake's docs are the most approachable starting point for warehouse-specific concepts (Time Travel, zero-copy cloning) before tackling BigQuery or Redshift's more infrastructure-heavy documentation.

Snowflake Documentation

Field	Details
Author / Organization	Snowflake Inc.
Category	Cloud Data Warehouses
Difficulty	Intermediate → Advanced
Best For	Snowflake-specific SQL extensions and warehouse architecture concepts (virtual warehouses, Time Travel, Streams/Tasks)
Why It Is Recommended	Clear separation between "SQL you already know" and "Snowflake-specific extensions," which makes it fast to onboard from a standard SQL background
Key Topics Covered	Virtual warehouses, semi-structured data (VARIANT), Time Travel, Streams and Tasks, Snowpark
Estimated Time	2–3 hours for "Getting Started"
Official Website	docs.snowflake.com
Free or Paid	Free
Prerequisites	18_SQL_BUSINESS_CASE_STUDIES
Who Should Read It	Analytics/data engineers using or evaluating Snowflake
Related SQL Handbook Modules	18_SQL_BUSINESS_CASE_STUDIES

BigQuery Documentation

Field	Details
Author / Organization	Google Cloud
Category	Cloud Data Warehouses
Difficulty	Intermediate → Advanced
Best For	Serverless, distributed SQL at very large scale, and BigQuery's specific pricing/performance model
Why It Is Recommended	Google Cloud's docs are unusually clear about the cost implications of query patterns — directly relevant since BigQuery bills by data scanned, not compute time
Key Topics Covered	GoogleSQL dialect specifics, partitioning and clustering, slot-based pricing, BigQuery ML

Field	Details
Estimated Time	2–3 hours for "Quickstarts"
Official Website	cloud.google.com/bigquery/docs
Free or Paid	Free
Prerequisites	Data Engineer Roadmap basics
Who Should Read It	Data engineers on or evaluating the Google Cloud stack
Related SQL Handbook Modules	13_SET_OPERATORS, 16_QUERY_OPTIMIZATION

Amazon Redshift Documentation

Field	Details
Author / Organization	Amazon Web Services
Category	Cloud Data Warehouses
Difficulty	Intermediate → Advanced
Best For	Redshift's PostgreSQL-derived SQL dialect and its columnar, cluster-based architecture
Why It Is Recommended	Clearly documents where Redshift's SQL diverges from standard PostgreSQL (important, since it's built on Postgres but isn't a drop-in replacement)
Key Topics Covered	Distribution keys and sort keys, COPY/UNLOAD, workload management (WLM), Redshift Spectrum
Estimated Time	2–3 hours for "Getting Started"
Official Website	docs.aws.amazon.com/redshift/
Free or Paid	Free
Prerequisites	PostgreSQL Official Documentation, for comparison
Who Should Read It	Data engineers on or evaluating the AWS data stack
Related SQL Handbook Modules	13_SET_OPERATORS, 16_QUERY_OPTIMIZATION

Analytical & Embedded Engines

Recommended reading order: DuckDB first if you want a fast, local way to feel the difference between OLTP and OLAP query execution. Spark SQL once you're actually working with distributed, cluster-scale data.

DuckDB Documentation

Field	Details
Author / Organization	DuckDB Foundation
Category	Analytical & Embedded Engines
Difficulty	Beginner → Intermediate
Best For	Learning columnar, vectorized query execution hands-on, locally, with zero infrastructure setup
Why It Is Recommended	The fastest way to feel the practical difference between a row-store (Postgres/MySQL) and a column-store engine, since it runs embedded — no server, no cluster, just a file
Key Topics Covered	Columnar storage, vectorized execution, Parquet/CSV integration, SQL dialect compatibility notes
Estimated Time	1–2 hours
Official Website	duckdb.org/docs/
Free or Paid	Free
Prerequisites	01_FUNDAMENTALS
Who Should Read It	Anyone curious about OLAP engines without wanting to stand up a warehouse first
Related SQL Handbook Modules	16_QUERY_OPTIMIZATION

Apache Spark SQL Documentation

Field	Details
Author / Organization	Apache Software Foundation
Category	Analytical & Embedded Engines
Difficulty	Advanced
Best For	Distributed SQL execution across a cluster, and the Catalyst optimizer that plans it

Field	Details
Why It Is Recommended	Primary source for how Spark's Catalyst optimizer and Tungsten execution engine plan and run distributed SQL — necessary reading before treating Spark SQL like "regular SQL on more machines"
Key Topics Covered	DataFrame/SQL API, Catalyst query optimizer, partitioning strategy, joins in a distributed context
Estimated Time	3–4 hours for the SQL Programming Guide
Official Website	<code>spark.apache.org/docs/latest/sql-programming-guide.html</code>
Free or Paid	Free
Prerequisites	Data Engineer Roadmap, 16_QUERY_OPTIMIZATION
Who Should Read It	Data engineers working with distributed processing frameworks
Related SQL Handbook Modules	16_QUERY_OPTIMIZATION, 19_SQL_PROJECTS

Analytics Engineering & Tooling

dbt Documentation

Field	Details
Author / Organization	dbt Labs
Category	Analytics Engineering & Tooling
Difficulty	Intermediate
Best For	Modeling, testing, and documenting SQL transformations the way a modern analytics team actually works
Why It Is Recommended	The primary source for dbt's modeling conventions (staging → intermediate → marts), testing framework, and the Semantic Layer — the practical backbone of the Analytics Engineer Roadmap
Key Topics Covered	Models, tests, macros (Jinja), sources and seeds, the Semantic Layer, project structure conventions
Estimated Time	3–4 hours for the Quickstart
Official Website	<code>docs.getdbt.com</code>
Free or Paid	Free

Field	Details
Prerequisites	06_CTEs, Analytics Engineer Roadmap
Who Should Read It	Analytics engineers, or anyone on the Analytics Engineer Roadmap
Related SQL Handbook Modules	projects/nagpurlens , projects/olist

SQLAlchemy Documentation

Field	Details
Author / Organization	Michael Bayer and the SQLAlchemy project
Category	Analytics Engineering & Tooling
Difficulty	Intermediate → Advanced
Best For	Understanding how a Python ORM translates to the SQL underneath it — useful even if you write raw SQL, to know what an ORM is doing on your behalf
Why It Is Recommended	Unusually transparent about the generated SQL behind ORM operations (the docs actively show you the SQL each pattern produces), which builds intuition for spotting ORM-generated N+1 query problems
Key Topics Covered	Core expression language vs. ORM, session/transaction management, relationship loading strategies (lazy, eager)
Estimated Time	3–4 hours for SQLAlchemy Core basics
Official Website	docs.sqlalchemy.org
Free or Paid	Free
Prerequisites	03_JOINS, comfort with Python
Who Should Read It	Backend developers using Python who want to understand the SQL their ORM generates
Related SQL Handbook Modules	Backend Developer Roadmap

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